OR4023 (Optimization) Homework 6

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Question 1. Let f be a differentiable function that is being minimized with an optimization algorithm that uses an exact line search. Let $\mathbf{p}_k$ be the search direction at $\mathbf{x}_k$, show that $\nabla f(\mathbf{x}_{k+1})\mathbf{p}_k = 0$.

Solution. Let $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$, where α_k is the step size chosen by exact line search, i.e., it minimizes $f(\mathbf{x}_k + \alpha \mathbf{p}_k)$ with respect to α.

Define $\varphi(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{p}_k)$. Since α_k is the minimizer, we have

$$\varphi'(\alpha_k) = 0.$$

By the chain rule,

$$\varphi'(\alpha) = \nabla f(\mathbf{x}_k + \alpha \mathbf{p}_k)^T \mathbf{p}_k.$$

Evaluating at α_k,

$$\varphi'(\alpha_k) = \nabla f(\mathbf{x}_k + \alpha_k \mathbf{p}_k)^T \mathbf{p}_k = \nabla f(\mathbf{x}_{k+1})^T \mathbf{p}_k = 0.$$

Therefore,

$$\nabla f(\mathbf{x}_{k+1}) \cdot \mathbf{p}_k = 0.$$

This shows that the gradient at the new point is orthogonal to the previous search direction when using exact line search.

Question 2. Prove that if $f(x) = \frac{1}{2}\sum_{i=1}^{m} r_i(x)^2$, then

$$\nabla^2 f(x) = \sum_{i=1}^{m} \left[\nabla r_i(x) \nabla r_i(x)^T + r_i(x) \nabla^2 r_i(x)\right].$$

Solution. Let $f(x) = \frac{1}{2}\sum_{i=1}^{m} r_i(x)^2$. We want to compute the Hessian $\nabla^2 f(x)$.

First, compute the gradient:

$$\nabla f(x) = \frac{1}{2} \sum_{i=1}^{m} \nabla \left(r_i(x)^2\right) = \sum_{i=1}^{m} r_i(x) \nabla r_i(x)$$

where we used the chain rule: $\nabla(r_i(x)^2) = 2 r_i(x) \nabla r_i(x)$, and the $1/2$ cancels the 2.

Now, compute the Hessian by differentiating the gradient:

$$\nabla^2 f(x) = \nabla \left(\sum_{i=1}^{m} r_i(x) \nabla r_i(x) \right) = \sum_{i=1}^{m} \nabla \left(r_i(x) \nabla r_i(x) \right)$$

For each i, apply the product rule:

$$\nabla \left(r_i(x) \nabla r_i(x) \right) = \nabla r_i(x) \nabla r_i(x)^T + r_i(x) \nabla^2 r_i(x)$$

where $\nabla r_i(x) \nabla r_i(x)^T$ is the outer product (a matrix), and $\nabla^2 r_i(x)$ is the Hessian of $r_i(x)$.

Therefore,

$$\nabla^2 f(x) = \sum_{i=1}^{m} \left[\nabla r_i(x) \nabla r_i(x)^T + r_i(x) \nabla^2 r_i(x)\right]$$

as required.

Question 3. Given data points $(t_i, y_i)_{i=1}^m$, determine the least squares fit by the function

$$y = x_1 + x_2 \cos t + x_3 \sin t$$

by finding the normal equation. Then find the least squares solution for the following data:

t_i	0	1	2	3	4
y_i	1.4	1.6	1.2	0.6	0.3

and draw the data points and the fitted curve.

Solution. The least squares problem is to minimize

$$f(x_1, x_2, x_3) = \sum_{i=1}^{m} \left(y_i - x_1 - x_2 \cos t_i - x_3 \sin t_i\right)^2 .$$

This is a linear least squares problem in the variables x_1, x_2, x_3.

Let us write it in matrix form. Define

$$A = \begin{bmatrix} 1 & \cos t_1 & \sin t_1 \\ 1 & \cos t_2 & \sin t_2 \\ \vdots & \vdots & \vdots \\ 1 & \cos t_m & \sin t_m \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

Then the normal equations are

$$A^T A\mathbf{x} = A^T \mathbf{y}$$

Now, for the given data:

$$A = \begin{bmatrix} 1 & 1.0000 & 0.0000 \\ 1 & 0.5403 & 0.8415 \\ 1 & -0.4161 & 0.9093 \\ 1 & -0.9900 & 0.1411 \\ 1 & -0.6536 & -0.7568 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1.4 \\ 1.6 \\ 1.2 \\ 0.6 \\ 0.3 \end{bmatrix}$$

Now, compute $A^T A$ and $A^T \mathbf{y}$:

$$A^T A = \begin{bmatrix} 5 & -0.5195 & 1.1351 \\ -0.5194 & 2.8724 & 0.4312 \\ 1.1351 & 0.4312 & 2.1276 \end{bmatrix}, \quad A^T \mathbf{y} = \begin{bmatrix} 5.1 \\ 0.9750 \\ 2.2951 \end{bmatrix}$$

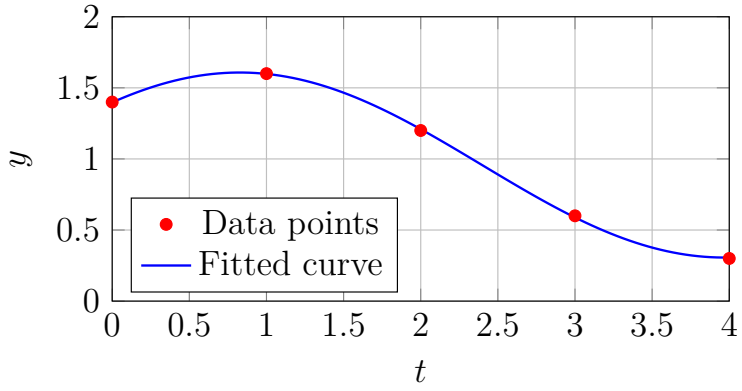
Solving the normal equation, we get

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \approx \begin{bmatrix} 1.025 \\ 0.393 \\ 0.730 \end{bmatrix}$$

So the least squares fit is:

$$y = 0.9571 + 0.4406\cos t + 0.4788\sin t$$

Plot: To draw the data points and the fitted curve, plot the points (t_i, y_i) for $t_i = 0, 1, 2, 3, 4$ and plot the curve $y = 0.9571 + 0.4406\cos t + 0.4788\sin t$ for t in $[0, 4]$. We see this is a small residual problem.



Question 4. Consider the least squares problem of fitting the data points

t_i	1	2	4	5	8
y_i	3.3	4.3	7.2	9	20

by the model $y = x_1 e^{x_2 t}$. Conduct two iterations of

(a) the classical Newton's method

(b) the Gauss Newton method

Solution. The function to minimize is

$$f(x) = \sum_{i=1}^{m} r_i^2(x).$$

where $r_i(x) = x_1 e^{x_2 t_i} - y_i$. Then

$$R(x) = \begin{bmatrix} r_1(x) \\ r_2(x) \\ \vdots \\ r_m(x) \end{bmatrix}, \quad \nabla R(x) = \begin{bmatrix} \nabla r_1(x) & \nabla r_2(x) & \cdots & \nabla r_m(x) \end{bmatrix}, \quad J(x) = \nabla R(x)^T.$$

$$\nabla f(x) = J(x)^T R(x), \quad \nabla^2 f(x) = J(x)^T J(x) + \sum_{i=1}^{m} r_i(x)\nabla^2 r_i(x).$$

(a) Classical Newton's method. Choose an initial guess $x_0 = (2.5, 0.25)^T$. At this point, we calculate

$$R(x_0) = \begin{bmatrix} -0.0899 \\ -0.1782 \\ -0.4043 \\ -0.2741 \\ -1.5274 \end{bmatrix}, \quad J(x_0) = \begin{bmatrix} 1.2840 & 3.2101 \\ 1.6487 & 8.2436 \\ 2.7183 & 27.1828 \\ 3.4903 & 43.6293 \\ 7.3891 & 147.7811 \end{bmatrix}$$

Hence

$$\nabla f(x_0) = J(x_0)^T R(x_0) = \begin{bmatrix} -13.7509 \\ -250.4232 \end{bmatrix}, \quad J^T(x_0)J(x_0) = \begin{bmatrix} 79 & 1336 \\ 1336 & 24560 \end{bmatrix}.$$

We have $\nabla^2 r_i(x_0) = \begin{bmatrix} 0 & t_i e^{x_2 t_i} \\ t_i e^{x_2 t_i} & t_i^2 x_1 e^{x_2 t_i} \end{bmatrix}$, So

$$\nabla^2 r_1(x_0) = \begin{bmatrix} 0 & 1.2840 \\ 1.2840 & 3.2101 \end{bmatrix}, \quad \nabla^2 r_2(x_0) = \begin{bmatrix} 0 & 3.2974 \\ 3.2974 & 16.4872 \end{bmatrix},$$

$$\nabla^2 r_3(x_0) = \begin{bmatrix} 0 & 10.8731 \\ 10.8731 & 108.7313 \end{bmatrix}, \quad \nabla^2 r_4(x_0) = \begin{bmatrix} 0 & 17.4517 \\ 17.4517 & 218.1464 \end{bmatrix}, \quad \nabla^2 r_5(x_0) = \begin{bmatrix} 0 & 59.1 \\ 59.1 & 1182 \end{bmatrix}.$$

So,

$$\nabla^2 f(x_0) = J(x_0)^T J(x_0) + \sum_{i=1}^{m} r_i(x_0) \nabla^2 r_i(x_0) = \begin{bmatrix} 79 & 1236 \\ 1236 & 22647 \end{bmatrix}.$$

The search direction is

$$\mathbf{p}_0 = -\nabla^2 f(x_0)^{-1} \nabla f(x_0) = \begin{bmatrix} 0.0079 \\ 0.0106 \end{bmatrix},$$

and the new iterate is

$$x_1 = x_0 + \mathbf{p}_0 = \begin{bmatrix} 2.5079 \\ 0.2606 \end{bmatrix}.$$

Repeat the above steps to get the second iterate.

(b) Gauss Newton method. For this method, the only difference is that we use $J(x)^T J(x)$ as the Hessian approximation. We have

$$J(x_0)^T J(x_0) = \begin{bmatrix} 79 & 1336 \\ 1336 & 24560 \end{bmatrix}.$$

The search direction is

$$\mathbf{p}_0 = -[J(x_0)^T J(x_0)]^{-1} \nabla f(x_0) = \begin{bmatrix} 0.0221 \\ 0.0090 \end{bmatrix},$$

and the new iterate is

$$x_1 = x_0 + \mathbf{p}_0 = \begin{bmatrix} 2.5221 \\ 0.2590 \end{bmatrix}.$$

Repeat the above steps to get the second iterate.

Question 5. Consider the constrained problem

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) = x_1^2 + x_1^2 x_3^2 + 2x_1 x_2 + x_2^4 + 8x_2 \\ \text{subject to} \quad & 2x_1 + 5x_2 + x_3 = 3 \end{aligned}$$

Determine which of the following points are stationary points: (i) $(0, 0, 2)^T$; (ii) $(0, 0, 3)^T$; (iii) $(1, 0, 1)^T$.

Solution. Let us check which of the given points are stationary points for the constrained problem.

Let $f(\mathbf{x}) = x_1^2 + x_1^2 x_3^2 + 2x_1 x_2 + x_2^4 + 8x_2$ and the constraint $g(\mathbf{x}) = 2x_1 + 5x_2 + x_3 - 3 = 0$.

A point $\mathbf{x}^*$ is a stationary point if there exists λ such that

$$\nabla f(\mathbf{x}^*) + \lambda \nabla g(\mathbf{x}^*) = 0$$

and $g(\mathbf{x}^*) = 0$.

Let us compute the gradients:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2x_1 + 2x_1 x_3^2 + 2x_2 \\ 2x_1 + 4x_2^3 + 8 \\ 2x_1^2 x_3 \end{bmatrix}$$

$$\nabla g(\mathbf{x}) = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

Now, check each point:

(i) $\mathbf{x}^{(1)} = (0, 0, 2)^T$

First, check the constraint:

$$g(\mathbf{x}^{(1)}) = 2 \cdot 0 + 5 \cdot 0 + 2 - 3 = -1 \neq 0$$

So this point does **not** satisfy the constraint and cannot be a stationary point.

(ii) $\mathbf{x}^{(2)} = (0, 0, 3)^T$

Constraint:

$$g(\mathbf{x}^{(2)}) = 2 \cdot 0 + 5 \cdot 0 + 3 - 3 = 0$$

So the constraint is satisfied.

Compute $\nabla f(\mathbf{x}^{(2)})$:

$$\begin{aligned} & x_1 = 0,\ x_2 = 0,\ x_3 = 3 \\ \frac{\partial f}{\partial x_1} &= 2 \cdot 0 + 2 \cdot 0 \cdot 9 + 2 \cdot 0 = 0 \\ \frac{\partial f}{\partial x_2} &= 2 \cdot 0 + 4 \cdot 0^3 + 8 = 8 \\ \frac{\partial f}{\partial x_3} &= 2 \cdot 0^2 \cdot 3 = 0 \end{aligned}$$

So $\nabla f(\mathbf{x}^{(2)}) = \begin{bmatrix} 0 \\ 8 \\ 0 \end{bmatrix}$.

We seek λ such that

$$\begin{bmatrix}0\\8\\0\end{bmatrix} + \lambda \begin{bmatrix}2\\5\\1\end{bmatrix} = 0 \implies \begin{cases}2\lambda = 0\\8 + 5\lambda = 0\\\lambda = 0\end{cases}$$

From the first equation, $\lambda = 0$. Plug into the second: $8 + 0 = 8 \neq 0$. Contradiction.

So there is **no** λ such that the KKT conditions are satisfied. Thus, $(0, 0, 3)^T$ is **not** a stationary point.

(iii) $\mathbf{x}^{(3)} = (1, 0, 1)^T$

Constraint:

$$g(\mathbf{x}^{(3)}) = 2 \cdot 1 + 5 \cdot 0 + 1 - 3 = 2 + 0 + 1 - 3 = 0$$

So the constraint is satisfied.

Compute $\nabla f(\mathbf{x}^{(3)})$:

$$x_1 = 1,\ x_2 = 0,\ x_3 = 1$$

$$\frac{\partial f}{\partial x_1} = 2 \cdot 1 + 2 \cdot 1 \cdot 1^2 + 2 \cdot 0 = 2 + 2 + 0 = 4$$

$$\frac{\partial f}{\partial x_2} = 2 \cdot 1 + 4 \cdot 0^3 + 8 = 2 + 0 + 8 = 10$$

$$\frac{\partial f}{\partial x_3} = 2 \cdot 1^2 \cdot 1 = 2$$

So $\nabla f(\mathbf{x}^{(3)}) = \begin{bmatrix}4\\10\\2\end{bmatrix}$.

We seek λ such that

$$\begin{bmatrix}4\\10\\2\end{bmatrix} + \lambda \begin{bmatrix}2\\5\\1\end{bmatrix} = 0 \implies \begin{cases}4 + 2\lambda = 0\\10 + 5\lambda = 0\\2 + \lambda = 0\end{cases}$$

From the third equation: $\lambda = -2$.

Plug into the first: $4 + 2(-2) = 4 - 4 = 0$ (OK).

Plug into the second: $10 + 5(-2) = 10 - 10 = 0$ (OK).

Therefore, there exists $\lambda = -2$ such that the KKT conditions are satisfied.

Conclusion:

- (i) $(0, 0, 2)^T$: **Not a stationary point** (does not satisfy the constraint)
- (ii) $(0, 0, 3)^T$: **Not a stationary point** (no λ solves KKT)
- (iii) $(1, 0, 1)^T$: **Is a stationary point** (KKT conditions satisfied with $\lambda = -2$)

Question 6. Use the method of Lagrangian multipliers to solve

$$\begin{aligned}&\text{minimize} && f(\mathbf{x}) = x_1^3 + x_2^3\\&\text{subject to} && 2x_1 + x_2 = 1\end{aligned}$$

Solution. Let $f(\mathbf{x}) = x_1^3 + x_2^3$ and the constraint $h(\mathbf{x}) = 2x_1 + x_2 - 1 = 0$.

Form the Lagrangian:

$$L(x_1, x_2, \lambda) = x_1^3 + x_2^3 + \lambda(2x_1 + x_2 - 1)$$

Set the gradients to zero:

$$\frac{\partial L}{\partial x_1} = 3x_1^2 + 2\lambda = 0 \tag{1}$$

$$\frac{\partial L}{\partial x_2} = 3x_2^2 + \lambda = 0 \tag{2}$$

$$\frac{\partial L}{\partial \lambda} = 2x_1 + x_2 - 1 = 0 \tag{3}$$

From (2): $\lambda = -3x_2^2$.

Substitute into (1):

$$3x_1^2 + 2(-3x_2^2) = 0 \implies 3x_1^2 - 6x_2^2 = 0 \implies x_1^2 = 2x_2^2$$

So $x_1 = \sqrt{2}x_2$ or $x_1 = -\sqrt{2}x_2$.

Now use the constraint (3):

$$2x_1 + x_2 = 1$$

Case 1: $x_1 = \sqrt{2}x_2$

$$2(\sqrt{2}x_2) + x_2 = 1 \implies (2\sqrt{2} + 1)x_2 = 1 \implies x_2 = \frac{1}{2\sqrt{2} + 1}$$

$$x_1 = \sqrt{2}x_2 = \frac{\sqrt{2}}{2\sqrt{2} + 1}$$

Case 2: $x_1 = -\sqrt{2}x_2$

$$2(-\sqrt{2}x_2) + x_2 = 1 \implies (-2\sqrt{2} + 1)x_2 = 1 \implies x_2 = \frac{1}{1 - 2\sqrt{2}}$$

$$x_1 = -\sqrt{2}x_2 = \frac{-\sqrt{2}}{1 - 2\sqrt{2}}$$

Now, check the second order condition:

$$\nabla_{xx}^2 L(\mathbf{x}) = \begin{bmatrix} 6x_1 & 0 \\ 0 & 6x_2 \end{bmatrix}.$$

For case 1, $x_1 > 0, x_2 > 0$, so $\nabla_{xx}^2 L(\mathbf{x}^{(1)})$ is positive definite, and the sufficient condition is satisfied. So the give point is a local minimum.

For case 2, $x_1 > 0, x_2 < 0$, so $\nabla_{xx}^2 L(\mathbf{x})$ is not positive definite. Now $\nabla h(x) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Let y be such that $\nabla h(x)^T y = 0$. Then $2y_1 + y_2 = 0$. So $y_2 = -2y_1$. For such y with $y \neq 0$, we have

$$y^T \nabla_{xx}^2 L(\mathbf{x}) y = 6x_1 y_1^2 + 6x_2 y_2^2 = 6x_1 y_1^2 + 24x_2 y_1^2 = (6x_1 + 24x_2)y_1^2 < 0,$$

so the point is not a local minimum, but a local maximum.